

Chapter 36. RV32/64G Instruction Set Listings

One goal of the RISC-V project is that it be used as a stable software development target. For this purpose, we define a combination of a base ISA (RV32I or RV64I) plus selected standard extensions (IMAFD, Zicsr, Zifencei) as a "general-purpose" ISA, and we use the abbreviation G for the IMAFDZicsr_Zifencei combination of instruction-set extensions. This chapter presents opcode maps and instruction-set listings for RV32G and RV64G.

Table 72. RISC-V base opcode map, $inst[1:0]=11$

$inst[4:2]$	000	001	010	011	100	101	110	$111 (>32b)$
$inst[6:5]$								
00	LOAD	LOAD-FP	<i>custom-0</i>	MISC-MEM	OP-IMM	AUIPC	OP-IMM-32	<i>reserved</i>
01	STORE	STORE-FP	<i>custom-1</i>	AMO	OP	LUI	OP-32	<i>reserved</i>
10	MADD	MSUB	NMSUB	NMADD	OP-FP	OP-V	<i>custom-2</i>	<i>reserved</i>
11	BRANCH	JALR	<i>reserved</i>	JAL	SYSTEM	OP-VE	<i>custom-3</i>	<i>reserved</i>

Table 72 shows a map of the major opcodes for RVG. Opcodes marked as *reserved* should be avoided for custom instruction-set extensions as they might be used by future standard extensions. Major opcodes marked as *custom-0* through *custom-3* will be avoided by future standard extensions and are recommended for use by custom instruction-set extensions within the base 32-bit instruction format.

We believe RV32G and RV64G provide simple but complete instruction sets for a broad range of general-purpose computing. The optional compressed instruction set described in [Chapter 28](#) can be added (forming RV32GC and RV64GC) to improve performance, code size, and energy efficiency, though with some additional hardware complexity.

As we move beyond IMAFDC into further instruction-set extensions, the added instructions tend to be more domain-specific and only provide benefits to a restricted class of applications, e.g., for multimedia or security. Unlike most commercial ISAs, the RISC-V ISA design clearly separates the base ISA and broadly applicable standard extensions from these more specialized additions.

31	27	26	25	24	20	19	15	14	12	11		7	6		0	
funct7				rs2		rs1		funct3		rd			opcode		R-type	
imm[11:0]						rs1		funct3		rd			opcode		I-type	
imm[11:5]				rs2		rs1		funct3		imm[4:0]			opcode		S-type	
imm[12 10:5]				rs2		rs1		funct3		imm[4:1 11]			opcode		B-type	
imm[31:12]										rd			opcode		U-type	
imm[20 10:1 11 19:12]										rd			opcode		J-type	

RV32I Base Instruction Set

imm[31:12]				rd	0110111	LUI
imm[31:12]				rd	0010111	AUIPC
imm[20 10:1 11 19:12]				rd	1101111	JAL
imm[11:0]		rs1	000	rd	1100111	JALR
imm[12 10:5]	rs2	rs1	000	imm[4:1 11]	1100011	BEQ
imm[12 10:5]	rs2	rs1	001	imm[4:1 11]	1100011	BNE
imm[12 10:5]	rs2	rs1	100	imm[4:1 11]	1100011	BLT
imm[12 10:5]	rs2	rs1	101	imm[4:1 11]	1100011	BGE
imm[12 10:5]	rs2	rs1	110	imm[4:1 11]	1100011	BLTU
imm[12 10:5]	rs2	rs1	111	imm[4:1 11]	1100011	BGEU
imm[11:0]		rs1	000	rd	0000011	LB
imm[11:0]		rs1	001	rd	0000011	LH
imm[11:0]		rs1	010	rd	0000011	LW
imm[11:0]		rs1	100	rd	0000011	LBU
imm[11:0]		rs1	101	rd	0000011	LHU
imm[11:5]	rs2	rs1	000	imm[4:0]	0100011	SB
imm[11:5]	rs2	rs1	001	imm[4:0]	0100011	SH
imm[11:5]	rs2	rs1	010	imm[4:0]	0100011	SW
imm[11:0]		rs1	000	rd	0010011	ADDI
imm[11:0]		rs1	010	rd	0010011	SLTI
imm[11:0]		rs1	011	rd	0010011	SLTIU
imm[11:0]		rs1	100	rd	0010011	XORI
imm[11:0]		rs1	110	rd	0010011	ORI
imm[11:0]		rs1	111	rd	0010011	ANDI
0000000	shamt	rs1	001	rd	0010011	SLLI
0000000	shamt	rs1	101	rd	0010011	SRLI
0100000	shamt	rs1	101	rd	0010011	SRAI
0000000	rs2	rs1	000	rd	0110011	ADD
0100000	rs2	rs1	000	rd	0110011	SUB
0000000	rs2	rs1	001	rd	0110011	SLL

0000000		rs2	rs1	010	rd	0110011	SLT
0000000		rs2	rs1	011	rd	0110011	SLTU
0000000		rs2	rs1	100	rd	0110011	XOR
0000000		rs2	rs1	101	rd	0110011	SRL
0100000		rs2	rs1	101	rd	0110011	SRA
0000000		rs2	rs1	110	rd	0110011	OR
0000000		rs2	rs1	111	rd	0110011	AND
fm	pred	succ	rs1	000	rd	0001111	FENCE
1000	0011	0011	00000	000	00000	0001111	FENCE.TSO
0000	0001	0000	00000	000	00000	0001111	PAUSE
000000000000			00000	000	00000	1110011	ECALL
000000000001			00000	000	00000	1110011	EBREAK

31	27	26	25	24	20	19	15	14	12	11	7	6	0	
funct7				rs2		rs1		funct3		rd		opcode		R-type
imm[11:0]						rs1		funct3		rd		opcode		I-type
imm[11:5]				rs2		rs1		funct3		imm[4:0]		opcode		S-type

RV64I Base Instruction Set (in addition to RV32I)

imm[11:0]		rs1	110	rd	0000011	LWU	
imm[11:0]		rs1	011	rd	0000011	LD	
imm[11:5]		rs2	rs1	011	imm[4:0]	SD	
000000	shamt		rs1	001	rd	0010011	SLLI
000000	shamt		rs1	101	rd	0010011	SRLI
010000	shamt		rs1	101	rd	0010011	SRAI
imm[11:0]		rs1	000	rd	0011011	ADDIW	
0000000		shamt	rs1	001	rd	0011011	SLLIW
0000000		shamt	rs1	101	rd	0011011	SRLIW
0100000		shamt	rs1	101	rd	0011011	SRAIW
0000000		rs2	rs1	000	rd	0111011	ADDW
0100000		rs2	rs1	000	rd	0111011	SUBW
0000000		rs2	rs1	001	rd	0111011	SLLW
0000000		rs2	rs1	101	rd	0111011	SRLW
0100000		rs2	rs1	101	rd	0111011	SRAW

RV32/RV64 Zifencei Standard Extension

imm[11:0]				rs1		001		rd		0001111		FENCE.I	
-----------	--	--	--	-----	--	-----	--	----	--	---------	--	---------	--

RV32/RV64 Zicsr Standard Extension

csr				rs1		001		rd		1110011		CSRRW	
csr				rs1		010		rd		1110011		CSRRS	
csr				rs1		011		rd		1110011		CSRRC	
csr				uimm		101		rd		1110011		CSRRWI	
csr				uimm		110		rd		1110011		CSRRSI	
csr				uimm		111		rd		1110011		CSRRCI	

RV32M Standard Extension

0000001			rs2		rs1		000		rd		0110011		MUL	
0000001			rs2		rs1		001		rd		0110011		MULH	
0000001			rs2		rs1		010		rd		0110011		MULHSU	
0000001			rs2		rs1		011		rd		0110011		MULHU	
0000001			rs2		rs1		100		rd		0110011		DIV	
0000001			rs2		rs1		101		rd		0110011		DIVU	

0000001	rs2	rs1	110	rd	0110011	REM
0000001	rs2	rs1	111	rd	0110011	REMU

RV64M Standard Extension (in addition to RV32M)

0000001	rs2	rs1	000	rd	0111011	MULW
0000001	rs2	rs1	100	rd	0111011	DIVW
0000001	rs2	rs1	101	rd	0111011	DIVUW
0000001	rs2	rs1	110	rd	0111011	REMW
0000001	rs2	rs1	111	rd	0111011	REMUW

31	27	26	25	24	20	19	15	14	12	11	7	6	0	
funct7				rs2		rs1		funct3		rd		opcode		R-type

RV32A Standard Extension								
00010	aq	rl	00000	rs1	010	rd	0101111	LR.W
00011	aq	rl	rs2	rs1	010	rd	0101111	SC.W
00001	aq	rl	rs2	rs1	010	rd	0101111	AMOSWAP.W
00000	aq	rl	rs2	rs1	010	rd	0101111	AMOADD.W
00100	aq	rl	rs2	rs1	010	rd	0101111	AMOXOR.W
01100	aq	rl	rs2	rs1	010	rd	0101111	AMOAND.W
01000	aq	rl	rs2	rs1	010	rd	0101111	AMOOR.W
10000	aq	rl	rs2	rs1	010	rd	0101111	AMOMIN.W
10100	aq	rl	rs2	rs1	010	rd	0101111	AMOMAX.W
11000	aq	rl	rs2	rs1	010	rd	0101111	AMOMINU.W
11100	aq	rl	rs2	rs1	010	rd	0101111	AMOMAXU.W

RV64A Standard Extension (in addition to RV32A)								
00010	aq	rl	00000	rs1	011	rd	0101111	LR.D
00011	aq	rl	rs2	rs1	011	rd	0101111	SC.D
00001	aq	rl	rs2	rs1	011	rd	0101111	AMOSWAP.D
00000	aq	rl	rs2	rs1	011	rd	0101111	AMOADD.D
00100	aq	rl	rs2	rs1	011	rd	0101111	AMOXOR.D
01100	aq	rl	rs2	rs1	011	rd	0101111	AMOAND.D
01000	aq	rl	rs2	rs1	011	rd	0101111	AMOOR.D
10000	aq	rl	rs2	rs1	011	rd	0101111	AMOMIN.D
10100	aq	rl	rs2	rs1	011	rd	0101111	AMOMAX.D
11000	aq	rl	rs2	rs1	011	rd	0101111	AMOMINU.D
11100	aq	rl	rs2	rs1	011	rd	0101111	AMOMAXU.D

31	27	26	25	24	20	19	15	14	12	11	7	6	0	
funct7				rs2		rs1		funct3		rd		opcode		R-type
rs3		funct2		rs2		rs1		funct3		rd		opcode		R4-type
imm[11:0]						rs1		funct3		rd		opcode		I-type
imm[11:5]				rs2		rs1		funct3		imm[4:0]		opcode		S-type

RV32F Standard Extension

imm[11:0]				rs1	010	rd	0000111	FLW
imm[11:5]		rs2	rs1	010	imm[4:0]	0100111	FSW	
rs3	00	rs2	rs1	rm	rd	1000011	FMADD.S	
rs3	00	rs2	rs1	rm	rd	1000111	FMSUB.S	
rs3	00	rs2	rs1	rm	rd	1001011	FNMSUB.S	
rs3	00	rs2	rs1	rm	rd	1001111	FMADD.S	
0000000		rs2	rs1	rm	rd	1010011	FADD.S	
0000100		rs2	rs1	rm	rd	1010011	FSUB.S	
0001000		rs2	rs1	rm	rd	1010011	FMUL.S	
0001100		rs2	rs1	rm	rd	1010011	FDIV.S	
0101100		00000	rs1	rm	rd	1010011	FSQRT.S	
0010000		rs2	rs1	000	rd	1010011	FSGNJ.S	
0010000		rs2	rs1	001	rd	1010011	FSGNJN.S	
0010000		rs2	rs1	010	rd	1010011	FSGNJX.S	
0010100		rs2	rs1	000	rd	1010011	FMIN.S	
0010100		rs2	rs1	001	rd	1010011	FMAX.S	
1100000		00000	rs1	rm	rd	1010011	FCVT.W.S	
1100000		00001	rs1	rm	rd	1010011	FCVT.WU.S	
1110000		00000	rs1	000	rd	1010011	FMV.X.W	
1010000		rs2	rs1	010	rd	1010011	FEQ.S	
1010000		rs2	rs1	001	rd	1010011	FLT.S	
1010000		rs2	rs1	000	rd	1010011	FLE.S	
1110000		00000	rs1	001	rd	1010011	FCLASS.S	
1101000		00000	rs1	rm	rd	1010011	FCVT.S.W	
1101000		00001	rs1	rm	rd	1010011	FCVT.S.WU	
1111000		00000	rs1	000	rd	1010011	FMV.W.X	

RV64F Standard Extension (in addition to RV32F)

1100000		00010	rs1	rm	rd	1010011	FCVT.L.S
1100000		00011	rs1	rm	rd	1010011	FCVT.LU.S
1101000		00010	rs1	rm	rd	1010011	FCVT.S.L
1101000		00011	rs1	rm	rd	1010011	FCVT.S.LU

31	27	26	25	24	20	19	15	14	12	11	7	6	0	
funct7				rs2		rs1		funct3		rd		opcode		R-type
rs3		funct2		rs2		rs1		funct3		rd		opcode		R4-type
imm[11:0]						rs1		funct3		rd		opcode		I-type
imm[11:5]				rs2		rs1		funct3		imm[4:0]		opcode		S-type

RV32D Standard Extension

imm[11:0]				rs1	011	rd	0000111	FLD
imm[11:5]		rs2	rs1	011	imm[4:0]	0100111	FSD	
rs3	01	rs2	rs1	rm	rd	1000011	FMADD.D	
rs3	01	rs2	rs1	rm	rd	1000111	FMSUB.D	
rs3	01	rs2	rs1	rm	rd	1001011	FNMSUB.D	
rs3	01	rs2	rs1	rm	rd	1001111	FNMADD.D	
0000001		rs2	rs1	rm	rd	1010011	FADD.D	
0000101		rs2	rs1	rm	rd	1010011	FSUB.D	
0001001		rs2	rs1	rm	rd	1010011	FMUL.D	
0001101		rs2	rs1	rm	rd	1010011	FDIV.D	
0101101		00000	rs1	rm	rd	1010011	FSQRT.D	
0010001		rs2	rs1	000	rd	1010011	FSGNJ.D	
0010001		rs2	rs1	001	rd	1010011	FSGNJN.D	
0010001		rs2	rs1	010	rd	1010011	FSGNJX.D	
0010101		rs2	rs1	000	rd	1010011	FMIN.D	
0010101		rs2	rs1	001	rd	1010011	FMAX.D	
0100000		00001	rs1	rm	rd	1010011	FCVT.S.D	
0100001		00000	rs1	rm	rd	1010011	FCVT.D.S	
1010001		rs2	rs1	010	rd	1010011	FEQ.D	
1010001		rs2	rs1	001	rd	1010011	FLT.D	
1010001		rs2	rs1	000	rd	1010011	FLE.D	
1110001		00000	rs1	001	rd	1010011	FCLASS.D	
1100001		00000	rs1	rm	rd	1010011	FCVT.W.D	
1100001		00001	rs1	rm	rd	1010011	FCVT.WU.D	
1101001		00000	rs1	rm	rd	1010011	FCVT.D.W	
1101001		00001	rs1	rm	rd	1010011	FCVT.D.WU	

RV64D Standard Extension (in addition to RV32D)

1100001		00010	rs1	rm	rd	1010011	FCVT.L.D
1100001		00011	rs1	rm	rd	1010011	FCVT.LU.D
1110001		00000	rs1	000	rd	1010011	FMV.X.D
1101001		00010	rs1	rm	rd	1010011	FCVT.D.L
1101001		00011	rs1	rm	rd	1010011	FCVT.D.LU

1111001	00000	rs1	000	rd	1010011	FMV.D.X
---------	-------	-----	-----	----	---------	---------

31	27	26	25	24	20	19	15	14	12	11	7	6	0	
funct7				rs2		rs1		funct3		rd		opcode		R-type
rs3		funct2		rs2		rs1		funct3		rd		opcode		R4-type
imm[11:0]						rs1		funct3		rd		opcode		I-type
imm[11:5]				rs2		rs1		funct3		imm[4:0]		opcode		S-type

RV32Q Standard Extension							
imm[11:0]			rs1	100	rd	0000111	FLQ
imm[11:5]		rs2	rs1	100	imm[4:0]	0100111	FSQ
rs3	11	rs2	rs1	rm	rd	1000011	FMADD.Q
rs3	11	rs2	rs1	rm	rd	1000111	FMSUB.Q
rs3	11	rs2	rs1	rm	rd	1001011	FNMSUB.Q
rs3	11	rs2	rs1	rm	rd	1001111	FNMADD.Q
0000011		rs2	rs1	rm	rd	1010011	FADD.Q
0000111		rs2	rs1	rm	rd	1010011	FSUB.Q
0001011		rs2	rs1	rm	rd	1010011	FMUL.Q
0001111		rs2	rs1	rm	rd	1010011	FDIV.Q
0101111		00000	rs1	rm	rd	1010011	FSQRT.Q
0010011		rs2	rs1	000	rd	1010011	FSGNJ.Q
0010011		rs2	rs1	001	rd	1010011	FSGNJN.Q
0010011		rs2	rs1	010	rd	1010011	FSGNJX.Q
0010111		rs2	rs1	000	rd	1010011	FMIN.Q
0010111		rs2	rs1	001	rd	1010011	FMAX.Q
0100000		00011	rs1	rm	rd	1010011	FCVT.S.Q
0100011		00000	rs1	rm	rd	1010011	FCVT.Q.S
0100001		00011	rs1	rm	rd	1010011	FCVT.D.Q
0100011		00001	rs1	rm	rd	1010011	FCVT.Q.D
1010011		rs2	rs1	010	rd	1010011	FEQ.Q
1010011		rs2	rs1	001	rd	1010011	FLT.Q
1010011		rs2	rs1	000	rd	1010011	FLE.Q
1110011		00000	rs1	001	rd	1010011	FCLASS.Q
1100011		00000	rs1	rm	rd	1010011	FCVT.W.Q
1100011		00001	rs1	rm	rd	1010011	FCVT.WU.Q
1101011		00000	rs1	rm	rd	1010011	FCVT.Q.W
1101011		00001	rs1	rm	rd	1010011	FCVT.Q.WU

RV64Q Standard Extension (in addition to RV32Q)														
1100011				00010		rs1	rm	rd		1010011		FCVT.L.Q		
1100011				00011		rs1	rm	rd		1010011		FCVT.LU.Q		

1101011	00010	rs1	rm	rd	1010011	FCVT.Q.L
1101011	00011	rs1	rm	rd	1010011	FCVT.Q.LU

31	27	26	25	24	20	19	15	14	12	11	7	6	0	
funct7				rs2		rs1		funct3		rd		opcode		R-type
rs3		funct2		rs2		rs1		funct3		rd		opcode		R4-type
imm[11:0]						rs1		funct3		rd		opcode		I-type
imm[11:5]				rs2		rs1		funct3		imm[4:0]		opcode		S-type

RV32Zfh Standard Extension							
imm[11:0]			rs1	001	rd	0000111	FLH
imm[11:5]		rs2	rs1	001	imm[4:0]	0100111	FSH
rs3	10	rs2	rs1	rm	rd	1000011	FMADD.H
rs3	10	rs2	rs1	rm	rd	1000111	FMSUB.H
rs3	10	rs2	rs1	rm	rd	1001011	FNMSUB.H
rs3	10	rs2	rs1	rm	rd	1001111	FNMADD.H
0000010		rs2	rs1	rm	rd	1010011	FADD.H
0000110		rs2	rs1	rm	rd	1010011	FSUB.H
0001010		rs2	rs1	rm	rd	1010011	FMUL.H
0001110		rs2	rs1	rm	rd	1010011	FDIV.H
0101110		00000	rs1	rm	rd	1010011	FSQRT.H
0010010		rs2	rs1	000	rd	1010011	FSGNJ.H
0010010		rs2	rs1	001	rd	1010011	FSGNJJ.H
0010010		rs2	rs1	010	rd	1010011	FSGNJX.H
0010110		rs2	rs1	000	rd	1010011	FMIN.H
0010110		rs2	rs1	001	rd	1010011	FMAX.H
0100000		00010	rs1	rm	rd	1010011	FCVT.S.H
0100010		00000	rs1	rm	rd	1010011	FCVT.H.S
0100001		00010	rs1	rm	rd	1010011	FCVT.D.H
0100010		00001	rs1	rm	rd	1010011	FCVT.H.D
0100011		00010	rs1	rm	rd	1010011	FCVT.Q.H
0100010		00011	rs1	rm	rd	1010011	FCVT.H.Q
1010010		rs2	rs1	010	rd	1010011	FEQ.H
1010010		rs2	rs1	001	rd	1010011	FLT.H
1010010		rs2	rs1	000	rd	1010011	FLE.H
1110010		00000	rs1	001	rd	1010011	FCLASS.H
1100010		00000	rs1	rm	rd	1010011	FCVT.W.H
1100010		00001	rs1	rm	rd	1010011	FCVT.WU.H
1110010		00000	rs1	000	rd	1010011	FMV.X.H
1101010		00000	rs1	rm	rd	1010011	FCVT.H.W
1101010		00001	rs1	rm	rd	1010011	FCVT.H.WU
1111010		00000	rs1	000	rd	1010011	FMV.H.X

RV64Zfh Standard Extension (in addition to RV32Zfh)						
1100010	00010	rs1	rm	rd	1010011	FCVT.L.H
1100010	00011	rs1	rm	rd	1010011	FCVT.LU.H
1101010	00010	rs1	rm	rd	1010011	FCVT.H.L
1101010	00011	rs1	rm	rd	1010011	FCVT.H.LU

Zawrs Standard Extension						
000000001101	00000	000	00000	1110011	WRS.NTO	
000000011101	00000	000	00000	1110011	WRS.STO	

Table 73 lists the CSRs that have currently been allocated CSR addresses. The timers, counters, and floating-point CSRs are the only CSRs defined in this specification.

Table 73. RISC-V control and status register (CSR) address map.

Number	Privilege	Name	Description
Floating-Point Control and Status Registers			
0x001	Read write	fflags	Floating-Point Accrued Exceptions.
0x002	Read write	frm	Floating-Point Dynamic Rounding Mode.
0x003	Read write	fcsr	Floating-Point Control and Status Register (frm + fflags).
Counters and Timers			
0xC00	Read-only	cycle	Cycle counter for RDCYCLE instruction.
0xC01	Read-only	time	Timer for RDTIME instruction.
0xC02	Read-only	instret	Instructions-retired counter for RDINSTRET instruction.
0xC80	Read-only	cycleh	Upper 32 bits of cycle , RV32I only.
0xC81	Read-only	timeh	Upper 32 bits of time , RV32I only.
0xC82	Read-only	instreth	Upper 32 bits of instret , RV32I only.